

Roll No.

Total No. of Pages : 02

Total No. of Questions : 16

B.Sc. (IT) / BCA (Sem.-3)
PROGRAMMING IN PYTHON
Subject Code : UGCA-1914
M.Code : 78180

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and students have to attempt any FOUR questions.

SECTION-A

Write briefly :

- 1) Multiline statement
- 2) Boolean Value
- 3) Pseudo Code
- 4) Python continue statement
- 5) String slice
- 6) Boolean expression
- 7) Lists in Python
- 8) String lower function
- 9) Python pass statement
- 10) Define Iteration

SECTION-B

- 11) How to check whether a given number is Palindrome or not?
- 12) What is the difference between modify and copy operations performed in dictionary?
- 13) Are Python strings immutable? Justify your answer.
- 14) Write Python Program to Calculate Area and Perimeter of different Shapes using Polymorphism.
- 15) Define conditional structures. How many of them are supported in Python? Explain with example.
- 16) Differentiate between indexing and negative indexing in Tuple?

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SECTION - B

2. What is an exception? Illustrate the use of exception handling by writing a function named `readposint` that uses the input dialog to prompt the user for a positive integer and then checks the input to confirm that it meets the requirements. It should be able to handle inputs that cannot be converted to int, as well as negative ints, and edge cases (e.g. when the user closes the dialog, or does not enter anything at all.)
3. Explain the process of opening, reading and writing files in Python. You are given a file called `class_scores.txt`, where each line of the file contains a one-word username and a test score separated by-spaces, like below:

GWashington 83

JAdams 86

Write code that scans through the file, adds 5 points to each test score", and outputs the usernames and new test scores to a new file, `scores2.txt`.

4. Write a detailed note on the four collection data types in the Python programming language: list, tuple, set and dictionary. Write about their features and usage.
5. What is a module? How are modules created and used in Python? List some standard Python modules.
6. What are the various types of operators available in Python? Write a program that generates a random decimal number between 1 and 10 with two decimal places of accuracy. Examples are 1.23, 3.45, 9.80, and 5.00.
7. Explain various mechanisms of passing arguments to functions in Python. Discuss the behaviour of recursive functions with the help of example.

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SECTION-B

2. Explain the steps of installing Python. Discuss the features and limitations of Python programming language.
3. What are variables? What are the various data types Python variables can have? Write a program to swap two numbers using temporary variable.
4. Write a short note on conditional branching statements supported by Python. When should we use nested if statements? Illustrate your answer with the help of example.
5. What do you understand by the term arguments? How do we pass them to a function? *“Arguments may be passed in the form of expressions to the called function”*. Justify this statement with the help of example.
6. Write a detailed note on opening and closing files in Python. Write about various access modes. Also, discuss the attributes of the File Object.
7. Discuss the object oriented nature of Python. Explain how classes are designed and how objects are created?

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